

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

01

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Q1**D**D. $(x+1)(x-2)$

Choice D is correct. According to the table, when x is -1 or 2 , $p(x) = 0$. Therefore, two x -intercepts of the graph of p are $(-1, 0)$ and $(2, 0)$. Since $(-1, 0)$ and $(2, 0)$ are x -intercepts, it follows that $(x+1)$ and $(x-2)$ are factors of the polynomial equation. This is because when $x = -1$, the value of $x+1$ is 0. Similarly, when $x = 2$, the value of $x-2$ is 0. Therefore, the product $(x+1)(x-2)$ is a factor of the polynomial function p .

Choice A is incorrect. The factor $x-3$ corresponds to an x -intercept of $(3, 0)$, which isn't present in the table. Choice B is incorrect. The factor $x+3$ corresponds to an x -intercept of $(-3, 0)$, which isn't present in the table. Choice C is incorrect. The factors $x-1$ and $x+2$ correspond to x -intercepts $(1, 0)$ and $(-2, 0)$, respectively, which aren't present in the table.

Q2

B

B. II only

Choice B is correct. Functions f and g are both exponential functions with a base of 0.40. Since 0.40 is less than 1, functions f and g are both decreasing exponential functions. This means that $f(x)$ and $g(x)$ decrease as x increases. Since $f(x)$ and $g(x)$ decrease as x increases, the maximum value of each function occurs at the least value of x for which the function is defined. It's given that functions f and g are defined for $x \geq 0$. Therefore, the maximum value of each function occurs at $x = 0$. Substituting 0 for x in the equation defining f yields $f(0) = 330.4^{0+3}$, which is equivalent to $f(0) = 330.4^3$, or $f(0) = 2.112$. Therefore, the maximum value of f is 2.112. Since the equation $f(x) = 330.4^{x+3}$ doesn't display the value 2.112, the equation defining f doesn't display the maximum value of f . Substituting 0 for x in the equation defining g yields $g(0) = 330.160.4^{0-2}$, which can be rewritten as $g(0) = 330.16 \frac{1}{0.4^2}$, or $g(0) = 330.16 \frac{1}{0.16}$, which is equivalent to $g(0) = 33$. Therefore, the maximum value of g is 33. Since the equation $g(x) = 330.160.4^{x-2}$ displays the value 33, the equation defining g displays the maximum value of g . Thus, only equation II displays, as a constant or coefficient, the maximum value of the function it defines.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**A****A.** 0, 36

Choice A is correct. It's given that the table shows values of x and their corresponding values of gx , where $gx = \frac{fx}{x+3}$. It's also given that f is a linear function. It follows that an equation that defines f can be written in the form $fx = mx + b$, where m represents the slope and b represents the y -coordinate of the y -intercept 0, b of the graph of $y = fx$ in the xy -plane. The slope of the graph of $y = fx$ can be found using two points, x_1, y_1 and x_2, y_2 , that are on the graph of $y = fx$, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since the table shows values of x and their corresponding values of gx , substituting values of x and gx in the equation $gx = \frac{fx}{x+3}$ can be used to define function f . Using the first pair of values from the table, $x = -27$ and $gx = 3$, yields $3 = \frac{f-27}{-27+3}$, or $3 = \frac{f-27}{-24}$. Multiplying each side of this equation by -24 yields $-72 = f-27$, so the point $-27, -72$ is on the graph of $y = fx$. Using the second pair of values from the table, $x = -9$ and $gx = 0$, yields $0 = \frac{f-9}{-9+3}$, or $0 = \frac{f-9}{-6}$. Multiplying each side of this equation by -6 yields $0 = f-9$, so the point $-9, 0$ is on the graph of $y = fx$. Substituting $-27, -72$ and $-9, 0$ for x_1, y_1 and x_2, y_2 , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{0 - (-72)}{-9 - (-27)}$, or $m = 4$. Substituting 4 for m in the equation $fx = mx + b$ yields $fx = 4x + b$. Since $0 = f-9$, substituting -9 for x and 0 for fx in the equation $fx = 4x + b$ yields $0 = 4(-9) + b$, or $0 = -36 + b$. Adding 36 to both sides of this equation yields $36 = b$. It follows that 36 is the y -coordinate of the y -intercept 0, b of the graph of $y = fx$. Therefore, the y -intercept of the graph of $y = fx$ is 0, 36.

Choice B is incorrect. 12 is the y -coordinate of the y -intercept of the graph of $y = gx$.

Choice C is incorrect. 4 is the slope of the graph of $y = fx$.

Choice D is incorrect. -9 is the x -coordinate of the x -intercept of the graph of $y = fx$.

Q4**-7**

The correct answer is -7 . For a quadratic function defined by an equation of the form $fx = ax - h^2 + k$, where a , h , and k are constants and $a > 0$, the function reaches its minimum when $x = h$. In the given function, $a = 1$, $h = -7$, and $k = 4$. Therefore, the value of x for which fx reaches its minimum is -7 .

Q5**-13/2**

Accept also: -6.5

The correct answer is $-\frac{13}{2}$. The value of x for which fx reaches its minimum can be found by rewriting the given equation in the form $fx = x - h^2 + k$, where fx reaches its minimum, k , when the value of x is h . The given equation, $fx = x - 2x + 15$, can be rewritten as $fx = x^2 + 13x - 30$. By completing the square, this equation can be rewritten as $fx = x^2 + 13x + \frac{13^2}{2} - 30 - \frac{13^2}{2}$, which is equivalent to $fx = x + \frac{13^2}{2} - \frac{289}{4}$, or $fx = x - \frac{13^2}{2} - \frac{289}{4}$. Therefore, fx reaches its minimum when the value of x is $-\frac{13}{2}$. Note that $-13/2$ and -6.5 are examples of ways to enter a correct answer.

Alternate approach: The graph of $y = fx$ in the xy -plane is a parabola. The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $fx = x - 2x + 15$, it follows that the two x -intercepts of the graph of $y = fx$ in the xy -plane occur when $x = 2$ and $x = -15$, or at the points $2, 0$ and $-15, 0$. The midpoint between two points, x_1, y_1 and x_2, y_2 , is $\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$. Therefore, the midpoint between $2, 0$ and $-15, 0$ is $\frac{2 + (-15)}{2}, \frac{0 + 0}{2}$, or $-\frac{13}{2}, 0$. It follows that fx reaches its minimum when the value of x is $-\frac{13}{2}$. Note that $-13/2$ and -6.5 are examples of ways to enter a correct answer.

Q6**C****C.** $fx = 1282^{x-1}$

Choice C is correct. For the form of the function in choice C, $fx = 1281.6^{x-1}$, the value of $f1$ can be found as 1281.6^{1-1} , which is equivalent to 1281.6^0 , or 128. Therefore, $k = 128$, which is shown in $fx = 1281.6^{x-1}$ as the coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**-24**

The correct answer is -24. Since the graph passes through the point 0, -6, it follows that when the value of x is 0, the value of y is -6. Substituting 0 for x and -6 for y in the given equation yields $-6 = 2(0)^2 + b(0) + c$, or $-6 = c$. Therefore, the value of c is -6. Substituting -6 for c in the given equation yields $y = 2x^2 + bx - 6$. Since the graph passes through the point -1, -8, it follows that when the value of x is -1, the value of y is -8. Substituting -1 for x and -8 for y in the equation $y = 2x^2 + bx - 6$ yields $-8 = 2(-1)^2 + b(-1) - 6$, or $-8 = 2 - b - 6$, which is equivalent to $-8 = -4 - b$. Adding 4 to each side of this equation yields $-4 = -b$. Dividing each side of this equation by -1 yields $4 = b$. Since the value of b is 4 and the value of c is -6, it follows that the value of bc is $4(-6)$, or -24.

Alternate approach: The given equation represents a parabola in the xy -plane with a vertex at -1, -8. Therefore, the given equation, $y = 2x^2 + bx + c$, which is written in standard form, can be written in vertex form, $y = a(x - h)^2 + k$, where h, k is the vertex of the parabola and a is the value of the coefficient on the x^2 term when the equation is written in standard form. It follows that $a = 2$. Substituting 2 for a , -1 for h , and -8 for k in this equation yields $y = 2(x - (-1))^2 - 8$, or $y = 2(x + 1)^2 - 8$. Squaring the binomial on the right-hand side of this equation yields $y = 2(x^2 + 2x + 1) - 8$. Multiplying each term inside the parentheses on the right-hand side of this equation by 2 yields $y = 2x^2 + 4x + 2 - 8$, which is equivalent to $y = 2x^2 + 4x - 6$. From the given equation $y = 2x^2 + bx + c$, it follows that the value of b is 4 and the value of c is -6. Therefore, the value of bc is $4(-6)$, or -24.

Q8**B**

B. $gx = 1444^x$

Choice B is correct. It's given that $fx = 94^x$ and $gx = fx + 2$. Substituting $x + 2$ for x in $fx = 94^x$ gives $fx + 2 = 94^{x+2}$. Rewriting this equation using properties of exponents gives $fx + 2 = 94^x 4^2$, which is equivalent to $fx + 2 = 94^x 16$. Multiplying 9 and 16 in this equation gives $fx + 2 = 1444^x$. Since $gx = fx + 2$, $gx = 1444^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

D

D. $C = 402.9^t$

Choice D is correct. It's given that an exponential model estimates that the number of comments on an article increased by a fixed percentage at the end of each hour. Therefore, the model can be represented by an exponential equation of the form $C = Ka^t$, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and K and a are constants. It's also given that when the article was first featured on the home page of the news website, there were 40 comments on the article. This means that when $t = 0$, $C = 40$. Substituting 0 for t and 40 for C in the equation $C = Ka^t$ yields $40 = Ka^0$, or $40 = K$. It's also given that the number of comments on the article at the end of an hour had increased by 190% of the number of comments on the article at the end of the previous hour. Multiplying the percent increase by the number of comments on the article at the end of the previous hour yields the number of estimated additional comments the article has on its home page: $40 \frac{190}{100}$, or 76 comments. Thus, the estimated number of comments for the following hour is the sum of the comments from the end of the previous hour and the number of additional comments, which is $40 + 76$, or 116. This means that when $t = 1$, $C = 116$. Substituting 1 for t , 116 for C , and 40 for K in the equation $C = Ka^t$ yields $116 = 40a^1$, or $116 = 40a$. Dividing both sides of this equation by 40 yields $2.9 = a$. Substituting 40 for K and 2.9 for a in the equation $C = Ka^t$ yields $C = 402.9^t$. Thus, the equation that best represents this model is $C = 40(2.9)^t$.

Choice A is incorrect. This model represents a situation where the number of comments at the end of each hour increased by 19% of the number of comments at the end of the previous hour, rather than 190%.

Choice B is incorrect. This model represents a situation where the number of comments at the end of each hour increased by 90% of the number of comments at the end of the previous hour, rather than 190%.

Choice C is incorrect. This model represents a situation where the number of comments at the end of each hour was 19 times the number of comments at the end of the previous hour, rather than increasing by 190% of the number of comments at the end of the previous hour.

Q10

-2112

The correct answer is -2,112. It's given that f is a quadratic function. It follows that f can be defined by an equation of the form $fx = ax - h^2 + k$, where a , h , and k are constants. It's also given that the table shows three values of x and their corresponding values of y , where $y = fx + 4$. Substituting $ax - h^2 + k$ for fx in this equation yields $y = ax - h^2 + k + 4$. This equation represents a quadratic relationship between x and y , where $k + 4$ is either the maximum or the minimum value of y , which occurs when $x = h$. For quadratic relationships between x and y , the maximum or minimum value of y occurs at the value of x halfway between any two values of x that have the same corresponding value of y . The table shows that x -values of 21 and 25 correspond to the same y -value, -8. Since 23 is halfway between 21 and 25, the maximum or minimum value of y occurs at an x -value of 23. The table shows that when $x = 23$, $y = 8$. It follows that $h = 23$ and $k + 4 = 8$. Subtracting 4 from both sides of the equation $k + 4 = 8$ yields $k = 4$. Substituting 23 for h and 4 for k in the equation $y = ax - h^2 + k + 4$ yields $y = ax - 23^2 + 4 + 4$, or $y = ax - 23^2 + 8$. The value of a can be found by substituting any x -value and its corresponding y -value for x and y , respectively, in this equation. Substituting 25 for x and -8 for y in this equation yields $-8 = a25 - 23^2 + 8$, or $-8 = a2^2 + 8$. Subtracting 8 from both sides of this equation yields $-16 = a2^2$, or $-16 = 4a$. Dividing both sides of this equation by 4 yields $-4 = a$. Substituting -4 for a , 23 for h , and 4 for k in the equation $fx = ax - h^2 + k$ yields $fx = -4x - 23^2 + 4$. The y -intercept of the graph of $y = fx$ in the xy -plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $fx = -4x - 23^2 + 4$ yields $f0 = -40 - 23^2 + 4$, or $f0 = -4 \cdot 23^2 + 4$. This is equivalent to $f0 = -2,112$, so the y -intercept of the graph of $y = fx$ in the xy -plane is 0, -2,112. Thus, the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane is -2,112.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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